

From the Surface Code to qLDPC Codes

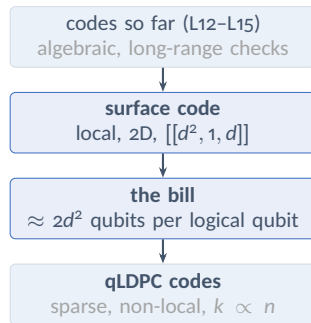
Local codes that work, and the codes that may replace them

Lecture 16

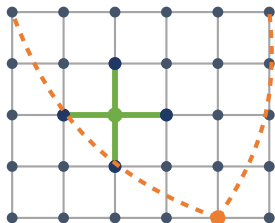
Term 261

Agenda

- 1 The surface code: a code that fits on a chip
- 2 Decoding, threshold, and what experiments show
- 3 The price of locality
- 4 qLDPC codes: hypergraph products, good codes, the gross code
- 5 The catches, and fault tolerance in one slide
- 6 Summary of the QEC block (L10–L16)



Why locality matters



green: local check (ancilla + 4 neighbours)

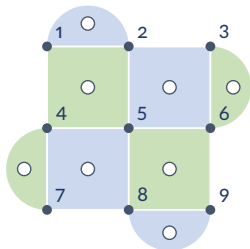
orange: non-local check, long wires

Measuring a check = an ancilla + one CNOT to each qubit in it. Every CNOT needs a physical **coupler**; on a chip, couplers join grid neighbours.

Code	weight	local at any d ?
Steane $[[7, 1, 3]]$	4	no
$[[5, 1, 3]]$	4	no
concatenated, large d	grows	no
surface code	4	yes

Takeaway: We want a code whose every check touches only neighbouring qubits on a plane — and whose distance still grows.

The surface code, distance 3



■ X check ■ Z check
● data ○ measure

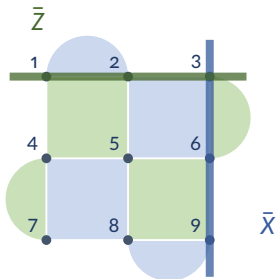
Data qubits on vertices; each face checks the qubits at its corners.

X checks	Z checks
X_1X_2, X_8X_9	Z_4Z_7, Z_3Z_6
$X_2X_3X_5X_6$	$Z_1Z_2Z_4Z_5$
$X_4X_5X_7X_8$	$Z_5Z_6Z_8Z_9$

- An X and a Z face share 0 or 2 qubits: all 8 commute
- 8 independent checks: $k = 9 - 8 = 1$
- Brute force over all 4^9 Paulis: no hidden logical of weight 1 or 2, so $d = 3$

Takeaway: $[[9, 1, 3]]$ on $9 + 8 = 17$ qubits. In general $[[d^2, 1, d]]$ on $2d^2 - 1$ qubits.

Logical operators are strings

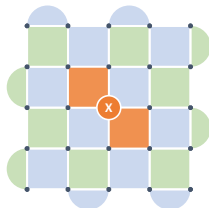


- $\bar{X} = X_3X_6X_9$ joins the **top and bottom** edges. It meets every Z face in 0 or 2 qubits: it commutes with all checks, yet is not a check
- $\bar{Z} = Z_1Z_2Z_3$ joins the **left and right** edges
- They share qubit 3, so they anticommute
- Strings deform: $X_1X_4X_7$ is $\bar{X}$ times checks

Distance = shortest edge-to-edge string

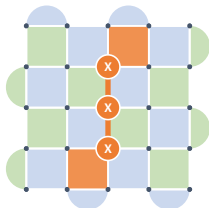
That string crosses the patch: d qubits. (Script, $d = 5$: the lightest logical has weight 5.)

Errors are chains; syndromes are their endpoints



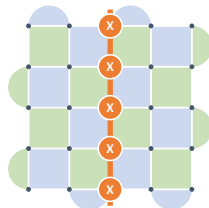
(a) one error

lights the 2 Z faces
it touches



(b) a chain of 3

inner faces see 2 errors:
only the ends light



(c) edge to edge

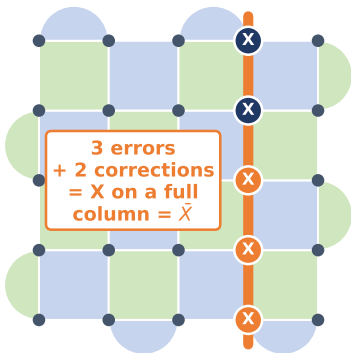
nothing lights:
it is $\bar{X}$

X errors light Z faces (**defects**); Z errors light X faces. A chain can also end silently on the top or bottom edge.

Takeaway: The decoder sees only endpoints. It must join them without completing an edge-to-edge string.

Watching a $d = 5$ patch decode

Cycle 4, step 4: apply the correction



What the decoder did, cycle by cycle

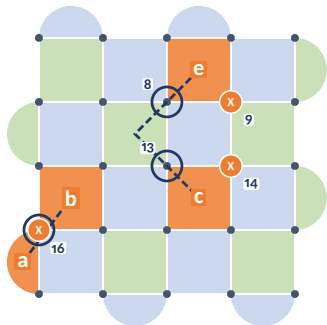
- 1 one error
two defects, paired ✓ **corrected**
- 2 two errors in a chain
defects only at the two ends ✓ **corrected**
- 3 one error at the edge
one defect, matched to the edge ✓ **corrected**
- 4 **three errors in a column**
one defect; the top edge is closer ✗ **logical error**

● X error ■ lit Z check - - matching ○ correction

$d = 5$ fixes any 2 errors. 3 errors along a column ($\lceil d/2 \rceil$) can fool matching: it closes the chain the short way.

Each cycle: errors → measure the checks → pair the lit checks → correct.

Decoding by minimum-weight perfect matching



X: true errors (hidden); **rings:** correction;
qubits numbered row by row from top left

Graph: nodes = lit checks (+ the edge); weight = fewest qubits between them. Choose the pairing of **least total weight**:

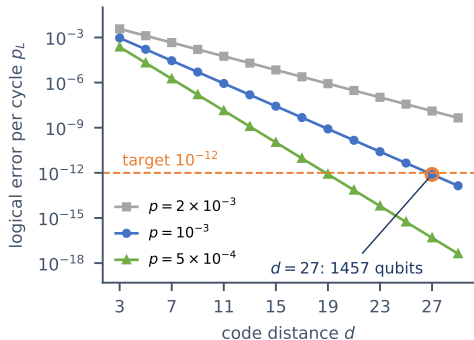
pairing of the defects	total weight
a-b, c-e	1 + 2 = 3
a-b, c-edge, e-edge	1 + 2 + 1 = 4
a-e, b-c	2 + 2 = 4
a-c, b-e	3 + 2 = 5

Correction $X_{16}X_{13}X_8$, true error $X_{16}X_{14}X_9$: different, but their product $X_8X_9X_{13}X_{14}$ **is a check**. Success (degeneracy, L13).

- Fails if error + correction spans the patch: $\lceil d/2 \rceil$ errors can do it
- Blossom algorithm: polynomial. **Union-find**: almost linear, $O(n \alpha(n))$

Dennis et al., J. Math. Phys. 43, 4452 (2002); Delfosse & Nickerson, Quantum 5, 595 (2021). Script: on $d = 5$, matching fixes all 325 X errors of weight ≤ 2 .

Threshold and scaling



Circuit-level threshold (noisy gates *and* measurements): **about 1%** (0.75% in 2007; 1.1–1.4% in 2011).

L13's formula with Fowler et al.'s fitted constants:

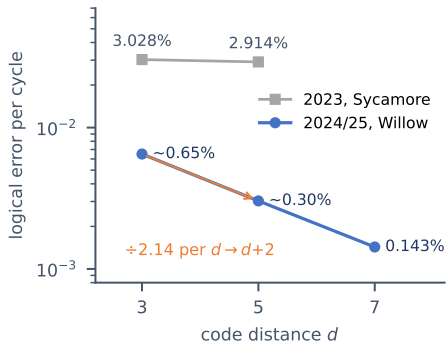
$$p_L \approx 0.03 (p/p_{\text{th}})^{(d+1)/2}, \quad p_{\text{th}} = 0.57\%$$

$p = 10^{-3}$, target p_L	10^{-6}	10^{-9}	10^{-12}
distance d	11	19	27
qubits $2d^2 - 1$	241	721	1457

Each $d \rightarrow d+2$ divides p_L by $\Lambda = p_{\text{th}}/p = 5.7$.

Fowler et al., PRA 86, 032324 (2012), Eq. 11 (p_L per cycle); Raussendorf & Harrington, PRL 98, 190504 (2007); Wang, Fowler & Hollenberg, PRA 83, 020302(R) (2011).

Experiments: below threshold (as of 2026)



2023, Sycamore: $d = 5$ (49 qubits) beat $d = 3$ (17), barely.
2024/25, Willow (105 qubits):

- $d = 3, 5, 7$: $\approx 0.65\%$, $\approx 0.30\%$, 0.143% error per cycle
- $\Lambda = 2.14 \pm 0.02$: each $d \rightarrow d+2$ halves p_L
- $d = 7$ (101 qubits) outlives the best physical qubit
 $2.4\times$: **beyond break-even**
- Real-time decoding at $d = 5$, $1.1\mu\text{s}$ cycle

Takeaway: At $\Lambda = 2.14$, reaching 10^{-12} needs $d = 63$ (7937 qubits per logical qubit); at $\Lambda = 5.7$, $d = 27$.

Google Quantum AI, Nature 614, 676 (2023); Google Quantum AI et al., Nature 638, 920 (2025), neural-network decoder.

The price of locality

Bravyi-Poulin-Terhal (2010)

Any code with local checks on a 2D lattice obeys $k d^2 = O(n)$.

The surface code saturates it: $k = 1$, $n \approx d^2$. In 2D, each logical qubit at distance d costs $\sim d^2$ qubits.

d	data d^2	total $2d^2 - 1$	rate k/n
5	25	49	0.040
11	121	241	0.0083
25	625	1249	0.0016

Worked example: 100 logical qubits at $d = 25$

$$100 \times (2 \cdot 25^2 - 1) = \mathbf{124,900}$$

physical qubits, before any gates or magic-state factories.

| *The rate goes to 0 as d grows. Which assumption must we drop to escape $k d^2 = O(n)$?*

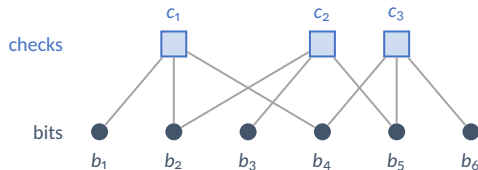
2D locality. Keep checks *sparse*; let them reach far.

Bravyi, Poulin, Terhal, "Tradeoffs for reliable quantum information storage in 2D systems," PRL 104, 050503 (2010).

Classical LDPC codes in one slide

A linear code $C = \ker H$ (L15) is **low-density parity-check** when H is sparse.

$$H = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{pmatrix}$$



Tanner graph: one edge per 1 in H

- Gallager (1962), rediscovered by MacKay and Neal (1996): near the Shannon limit
- Decoded by **belief propagation**: messages along the few edges, linear time; inside 5G and Wi-Fi today

Takeaway: Sparse checks wired as an expander, not a grid: constant rate *and* distance growing with n .

R. G. Gallager, IRE Trans. Inf. Theory 8, 21 (1962); D. J. C. MacKay & R. M. Neal, Electron. Lett. 32, 1645 (1996).

Quantum LDPC codes, and the obstruction

Definition

A **qLDPC code** is a CSS code (L15) whose H_X, H_Z are sparse: each check acts on $\leq w$ qubits, each qubit is in $\leq c$ checks, w, c **constant** as n grows.

- The surface code is qLDPC ($w = 4, c = 2$), but local, so stuck at $kd^2 = O(n)$
- Goal: $k \propto n$ with growing d , small w, c

The obstruction. X and Z checks must commute:

$$H_X H_Z^T = 0 \pmod{2}$$

Every X check overlaps every Z check in an *even* number of qubits.

Two good classical LDPC matrices picked independently will not satisfy this. For about two decades no qLDPC family had d much beyond $\sqrt{n}$.

Takeaway: qLDPC research is a search for *constructions* that make $H_X H_Z^T = 0$ automatic while keeping H_X, H_Z sparse.

The hypergraph product

Tillich-Zémor (2009). From any two classical codes $H_1 (m_1 \times n_1)$, $H_2 (m_2 \times n_2)$:

$$H_X = (H_1 \otimes \mathbb{I}_{n_2} \mid \mathbb{I}_{m_1} \otimes H_2^T), \quad H_Z = (\mathbb{I}_{n_1} \otimes H_2 \mid H_1^T \otimes \mathbb{I}_{m_2})$$

$H_X H_Z^T = H_1 \otimes H_2^T + H_1 \otimes H_2^T = 0$ for any H_1, H_2 . Sparse in, sparse out.

n	$n_1 n_2 + m_1 m_2$
k	$k_1 k_2 + k_1^T k_2^T$
d	$\min(d_1, d_2, d_1^T, d_2^T)$

T : the code $\ker H^T$ (none if H has full rank)

Checked in the script

- repetition₃ × repetition₃ = $[[13, 1, 3]]$: the **surface code** (unrotated)
- Hamming × Hamming = $[[58, 16, 3]]$: 16 logical qubits, checks of weight 5-7

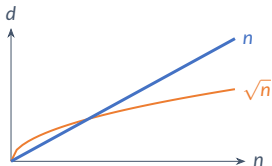
From good classical LDPC codes ($k_i, d_i \propto n_i$)

$k \propto n$ (constant rate), but $d \propto \sqrt{n}$: the same $\sqrt{n}$ as the surface code

J.-P. Tillich & G. Zémor, IEEE Trans. Inf. Theory 60, 1193 (2014); arXiv:0903.0566 (2009).

Breaking $\sqrt{n}$: good qLDPC codes exist

Family	k	d	2D local?
Surface code	1	$\sqrt{n}$	yes
Hypergraph product	$\propto n$	$\propto \sqrt{n}$	no
Good qLDPC	$\propto n$	$\propto n$	no



The catch: these are asymptotic, with large constants. At the sizes of the next decade, small hand-searched codes win (next slide).

- **Lifted products** (Panteleev–Kalachev, 2022): products over a group, not just $\otimes$
- **Quantum Tanner codes** (Leverrier–Zémor, 2022): small codes placed on an expander complex
- Constant check weight, $k = \Theta(n)$, $d = \Theta(n)$: the quantum twin of good classical LDPC codes

Bivariate bicycle codes

Qubits on an $\ell \times m$ torus, two per site; cyclic shifts x, y along the two directions commute.

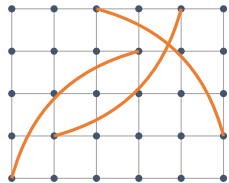
$$A = x^3 + y + y^2, \quad B = y^3 + x + x^2$$

$$H_X = [A \mid B], \quad H_Z = [B^T \mid A^T]$$

$$H_X H_Z^T = AB + BA = 0 \text{ because } AB = BA.$$

Script, $\ell = 12, m = 6$:

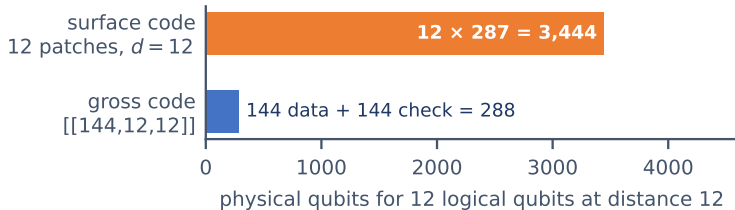
- $n = 2\ell m = 144, k = 12$ (from ranks); $d = 12$ (from the paper)
- every check has weight 6; every qubit is in 3 X and 3 Z checks



Wiring: degree 6, two edge-disjoint **planar** layers.
Most links short, some **long**.

S. Bravyi, A. W. Cross, J. M. Gambetta, D. Maslov, P. Rall, T. J. Yoder, Nature 627, 778 (2024).

The gross code $[[144, 12, 12]]$



Same distance (our count): 12 surface patches at $d = 12$ need $12 \times 287 = 3444$ qubits; the gross code needs 288: **$12 \times$ fewer.**

The paper's claim ($p = 0.1\%$): 288 qubits keep 12 logical qubits for nearly 10^6 cycles; the surface code would need **nearly 3000**. Threshold 0.7%.

Takeaway: 24 physical qubits per logical qubit instead of 287.

The catches

1. Connectivity

Long-range checks need long couplers. IBM's *Loon* chip (2025) adds on-chip "c-couplers" between distant qubits. Every long wire is a new error source.

2. Decoding

One error lights 3 checks, not 2: **no matching**.



- **BP+OSD**: accurate, cubic time
- **Relay-BP** (2025): built for FPGA/ASIC
- Budget: $\sim 1 \mu\text{s}$ per round

3. Logical gates

No lattice-surgery recipe. Gates come from code symmetries, ancilla "surgery" systems, or teleporting into a surface-code patch: active research.

Roffe et al., PRR 2, 043423 (2020); Müller et al., arXiv:2506.01779 (2025); IBM newsroom, 12 Nov. 2025 (Loon; claims real-time qLDPC decoding in $< 480 \text{ ns}$).

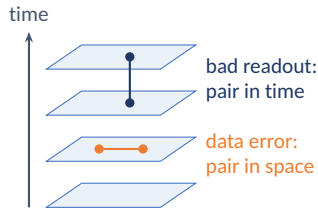
Platforms and roadmaps (as of 2026)

Platform	Fit for qLDPC	Status
Superconducting	fixed planar wiring; needs extra coupler layers	IBM roadmap: Loon (2025) → Kookaburra (2026) → Cockatoo (2027) → Starling (2029) : 200 logical qubits, 10^8 operations (a plan). [[18, 4, 4]] BB code run on a 32-qubit long-range chip (Kunlun)
Neutral atoms	atoms are <i>moved</i> : long range nearly free	proposal: rearrangement runs the checks; beats the surface code from a few hundred qubits (simulation)
Trapped ions	all-to-all within a trap	4 logical in 18 physical qubits at break-even (2026 preprint)

The surface code was designed *for* the planar chip; qLDPC codes ask the hardware to change.

IBM, roadmap 10 June 2025; K. Wang et al., Nat. Phys. (2026), arXiv:2505.09684; Q. Xu et al., Nat. Phys. 20, 1084 (2024); E. Tham et al., arXiv:2606.06455 (2026).

What fault tolerance adds, in one slide



- 1 **Measurements fail too.** Repeat the checks for $\sim d$ rounds; decode the space-time history (matching in 3D).
- 2 **Logical Cliffords** stay inside the code: *lattice surgery* merges and splits patches to measure $\bar{X}\bar{X}$ or $\bar{Z}\bar{Z}$; *transversal* gates where qubits can move.
- 3 **The T gate.** No code has a transversal universal gate set (Eastin–Knill). Distil noisy $|T\rangle$ states (15 in, 1 out, error $\approx 35p^3$) and teleport them in. These factories often dominate machine-size estimates.

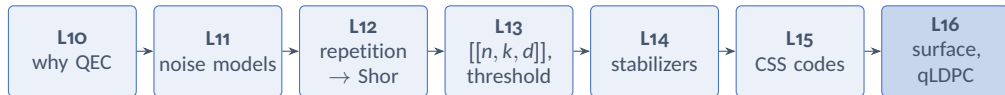
Dennis et al. (2002); Horsman et al., NJP 14, 123011 (2012); Eastin & Knill, PRL 102, 110502 (2009); Bravyi & Kitaev, PRA 71, 022316 (2005).

Surface vs. qLDPC at a glance

	Surface code	Hypergraph product	Bivariate bicycle
$[[n, k, d]]$	$[[d^2, 1, d]]$	$k \propto n, d \propto \sqrt{n}$	gross: $[[144, 12, 12]]$
Rate k/n	$1/d^2 \rightarrow 0$	constant	1/12
Check weight	4	bounded (5-7 in our example)	6
Connectivity	2D nearest neighbour	non-local	2 planar layers, some long
Decoder	matching, union-find	BP+OSD	BP+OSD, Relay-BP
Logical gates	lattice surgery	research	research
Demonstrated	below threshold, $d = 7$	not at scale	$[[18, 4, 4]]$; gross code not yet

Takeaway: The surface code is what works today. qLDPC codes promise $\sim 10\times$ fewer qubits, paid for in wiring, decoding and gates.

Summary of the QEC block, L10–L16



- Surface code $[[d^2, 1, d]]$: weight-4 local checks; errors are chains, syndromes their endpoints
- Matching pairs the endpoints; below $p_{\text{th}} \approx 1\%$, p_L drops by Λ per $d \rightarrow d+2$ (Willow: $\Lambda = 2.14$, $d = 7$ beyond break-even)
- Locality costs: $kd^2 = O(n)$ in 2D, so $\sim 2d^2$ qubits per logical qubit
- qLDPC: sparse, non-local checks. Hypergraph product ($k \propto n$, $d \propto \sqrt{n}$), good codes ($k, d \propto n$), gross code (12 logical qubits in 288)
- The bill moves to wiring, decoding and gates; fault tolerance adds repeated rounds and magic states

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Surface code

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Experiments (as of 2026)

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Fault tolerance and textbooks

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